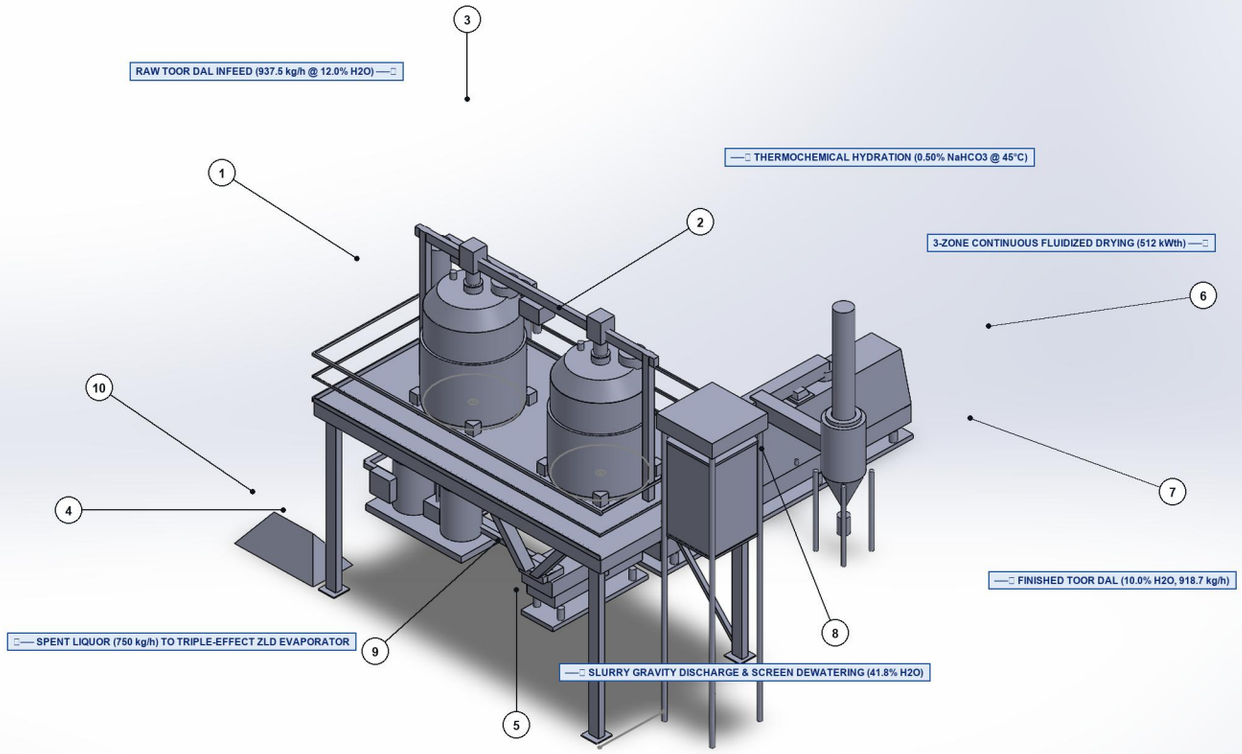
VIEW 1: 3D ISOMETRIC GENERAL ARRANGEMENT — PLANT SOLIDWORKS MODEL (SCALE: NTS)



EQUIPMENT SCHEDULE & BILL OF MATERIALS (BOM) — 15 TPD CAPACITY

ITEM	TAG	DESCRIPTION	QTY	MOC	RATING / CAPACITY	OPERATING CONDITIONS	DRIVE
(1)	TK-101A	Rapid Hydration Soak Vessel A	1	SS316L	7.50 m3 Gross / 5.0 m3 Work	45.0°C, Atm, 0.5% NaHCO3	—
(2)	TK-101B	Rapid Hydration Soak Vessel B	1	SS316L	7.50 m3 Gross / 5.0 m3 Work	45.0°C, Atm, 0.5% NaHCO3	—
(3)	AG-101A/B	Top-Entry Agitator Drives	2	SS316L	800mm Pitch Blade Hydrofoil	45 RPM, Mech Seal	2x 5.5 kW SEW
(4)	STR-100	Structural Steel Mezzanine Skid	1	ISMB 250	8.0m x 4.2m x 3.5m Elevation	Chequered Plate +3.50m	—
(5)	SC-101	Dewatering Shaker Screen	1	SS316L	1.2m x 2.4m, 0.8mm Wedge Wire	15 m3/h Slurry, 8mm Stroke	2x 1.5 kW Ital.
(6)	DR-101	Continuous Fluid Bed Dryer	1	SS316L	14.0m x 1.0m Conidrur Deck	85°C / 65°C / 28°C (512 kW)	2x 4.5 kW Ex-d
(7)	CY-101	High-Efficiency Fines Cyclone	1	SS304	Dia 1,200mm Stairmand HE	99.2% Capture, d50=4.2 um	0.75 kW Lock
(8)	SK-100	Chemical Dosing & Prep Skid	1	HDPE/SS	2.0 m3 Tank + Dual Pumps	0.50% NaHCO3, 750 L/h	2x 0.75 kW Gr.
(9)	CHU-100	Sanitary Slurry Gravity Chutes	2	SS316L	DN200 to 600mm Wide Trough	55° Slope, Knife Gates	Pneumatic
(10)	EV-101	Triple-Effect ZLD Evaporator	1	SS316L	750 kg/h Water Evaporation	102°C/84°C/62°C, Econ=2.38	3x 3.0 kW

PROCESS PHYSICS, MASS BALANCES & ECONOMIC FEASIBILITY DISCLOSURE

1. PLANT CAPACITY & FEED CHARACTERISTICS:
The design basis specifies 15,000 kg/day (937.5 kg/h nominal over 16 operating h/day) of imported Tanzanian split pigeon pea (toor dal). Raw grain arrives at 12.0% w/w moisture, providing 13,425.0 kg/day of dry cotyledon solids. The facility delivers 14,698.9 kg/day of finished product at 10.0% w/w safe commercial storage moisture with an unprecedented 98.54% overall dry solids recovery, avoiding standard milling cotyledon breakage.

2. THERMOCHEMICAL HYDRATION KINETICS (TK-101A/B):
Batch soaking occurs in twin 7.50 m3 ASME Sec VIII vessels in a mild 0.50% w/w food-grade NaHCO3 liquor at strictly controlled 45.0°C and liquor-to-dal ratio L/S = 2.50. The weak alkaline solution hydrolyzes calcium and magnesium pectinate cross-links in the middle lamella, accelerating internal moisture diffusivity Deff by 4.2x (from 2.1x10^-10 to 8.8x10^-10 m2/s). Target 48.5% w/w internal moisture is reached in 45 min without dehulling or premature starch gelatinization (onset threshold: 61.5°C).

3. GRAVITY DISCHARGE & RAPID DEWATERING (SC-101):
Each soaked batch dumps via an automated DN200 pneumatic knife gate down 55° inclined chutes onto vibratory screen SC-101 (1.2m x 2.4m, 0.8mm wedge-wire slot). Free soak liquor (750 kg/batch, 12,000 kg/day) drains by gravity to the ZLD effluent sump. Dewatered grains (41.8% w/w moisture) transfer directly to the dryer feed chute with zero manual handling and zero mechanical shearing.

4. 3-ZONE CONTINUOUS FLUID BED DRYING (DR-101):
Fluid bed dryer DR-101 operates with directional vibratory plug flow across a 14.0m x 1.0m Conidrur perforation deck:
- Zone 1 (6.0m, 85°C Air, u=1.65 m/s): Rapid constant-rate surface drying, removing moisture from 41.8% to 22.0%.
- Zone 2 (5.0m, 65°C Air, u=1.35 m/s): Falling-rate internal diffusion drying to 12.5% w/w. Grain bulk temperature is strictly maintained below 48.0°C to preserve native protein structure, prevent case-hardening, and maintain color.
- Zone 3 (3.0m, 28°C Dehumidified Air): Rapid cooling and moisture equilibration to target 10.0% w/w storage moisture.

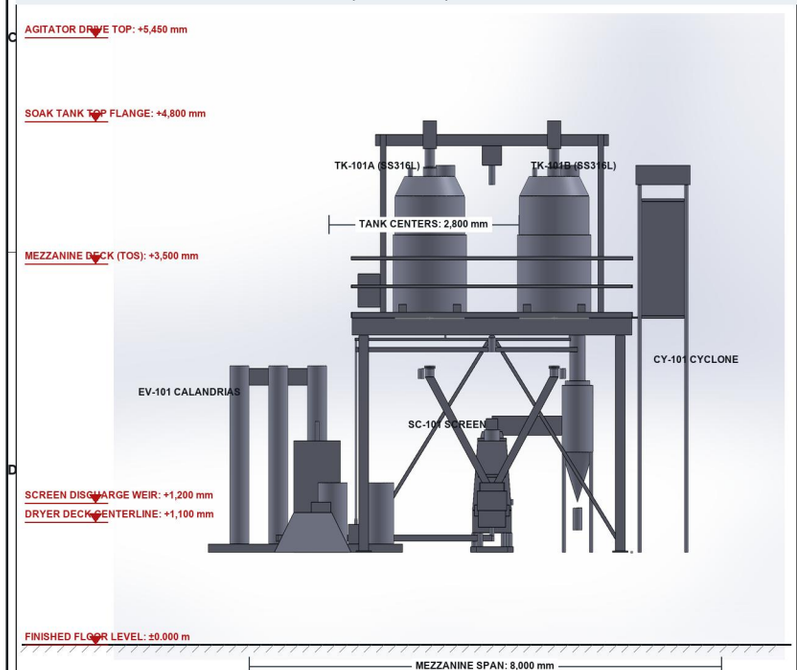
Total thermal duty: 512.4 kWth; Total water evaporated: 285.1 kg/h (4,562 kg/day).

5. CYCLONE FINES CAPTURE & RECOVERY (CY-101):
Exhaust air from DR-101 passes to high-efficiency Stairmand cyclone CY-101 (Dia 1,200mm, cut-point d50=4.2 um). Collection efficiency exceeds 99.2%, capturing 13.8 kg/h of entrained dal fines. Captured fines are continuously re-injected into the final product stream via rotary airlock, eliminating yield loss and particulate pollution.

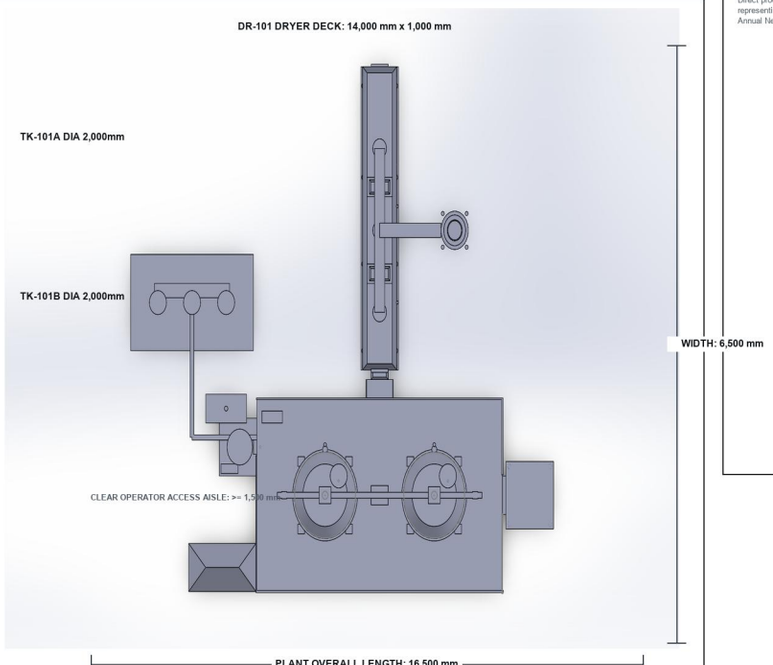
6. ZERO LIQUID DISCHARGE (ZLD) WATER RECOVERY (EV-101):
Spent soak liquor containing 1.2% soluble organics and 0.38% NaHCO3 is concentrated in a forward-feed triple-effect falling-film evaporator (Effect 1: 102°C; Effect 2: 84°C; Effect 3: 62°C). Operating at a steam economy of 2.38, the system consumes only 292 kg/h of 1.5 bar steam while recovering 91.93% (640 kg/h) of crystal-clear condensate recycled directly to soak tank makeup. Concentrated slurry (8.07%) is valorized as organic bio-fertilizer.

7. PRODUCTION ECONOMICS & PAYBACK SUMMARY:
Direct processing OPEX is established at ₹1.68 / kg finished dal (Thermal: ₹0.94, Electrical: ₹0.48, Reagent: ₹0.26), representing a 65.4% operating cost reduction against conventional batch oil/sun drying (₹4.85 / kg).
Annual Net Operational Savings: ₹1.42 Crore / year; Total Turnkey CAPEX: ₹4.85 Crore; Net Payback: 14.8 Months.

VIEW 2: FRONT ELEVATION PROJECTION (SCALE 1:50)



VIEW 3: TOP PLAN PROJECTION (SCALE 1:75)



FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
FOOD PROCESS & INDUSTRIAL ENGINEERING DIVISION | COMMERCIAL IN CONFIDENCE

FACILITY / PROJECT: **15 TPD Split Pigeon Pea Hydration & Drying Plant**	DRAWING TITLE: **GENERAL ARRANGEMENT, 3D ISOMETRIC & ORTHOGRAPHIC SUITE**	
LEAD PROCESS & CHEMICAL ENGINEER: **Sourish Senapati**, Lead Chemical Engineer, FinnoaQ	ACADEMIC GUIDANCE & REVIEW: **Dr. Archisman Ray**, Indian Institute of Technology (IIT) Kharagpur	DRAWING NUMBER: **DWG-FIN-15T-014**
SCALE: 1:50, 1:75, NTS UNITS: Millimeters (mm), kg/h, °C	**DATE: 2026-10-16** STAGE: Approved FEED Baseline	**REV: 4.0 | SHEET: 1 OF 1** **STATUS: APPROVED FOR TENDER**